

Safaa Mamoun

Neuroscientist Integrating Behaviour, Computation, and Neural Data

Trieste, Italy

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Research Focus

Computational and behavioural neuroscience, with a focus on behavioural dynamics, social behaviour, stress, aging, and the neural substrates of internal states.

Current Position

2022–present	PhD Researcher – Neurobiology <i>Scuola Internazionale Superiore di Studi Avanzati (SISSA), Trieste, Italy</i> • Investigating age-dependent effects of social isolation on anxiety-related behaviour, social behaviour, and neural plasticity in zebrafish. • Designing and implementing computational behavioural analysis pipelines using pose estimation, unsupervised behavioural segmentation, statistical modelling, and machine learning. • Analysing behavioural organization using entropy-based measures, factor analysis, transition matrices, and Markov-chain modelling. • Conducting histological staining and structural neural analyses, including dendritic spine quantification. • Project conducted in collaboration with Prof. Davide Zoccolan.
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Education

2022–present	PhD in Neurobiology <i>Expected defence: late 2026 International School for Advanced Studies (SISSA), Trieste, Italy</i> • Doctoral research combining experimental neurobiology, quantitative behavioural analysis, and computational modelling.
2016–2018	MSc in Genetics and Molecular Biology <i>University of Khartoum, Sudan</i> • Master's thesis: <i>Association of COMT Val158Met Polymorphism with Fibromyalgia in Khartoum State, Sudan.</i> • Training in molecular genetics, biomedical research methods, and human genetics data analysis.
2008–2013	BSc in Zoology <i>University of Khartoum, Sudan</i> • Bachelor's thesis: <i>Grivet Monkey Behavioural Adaptations in a University Campus Habitat.</i> • Training in zoology, physiology, biochemistry, behavioural observation, and field-based data collection.

Research Experience

Mar–Jun 2022	Visiting Researcher – Cognitive Neuroscience <i>SISSA, Trieste, Italy</i> • Participated in a collaborative project on incidental learning, integrating behavioural data analysis with theoretical perspectives on memory and learning. • Contributed to experimental-design discussions and interpretation of behavioural data. • Gained exposure to formal modelling and hypothesis-driven analysis under the supervision of Prof. Alessandro Treves.
2014–2015	Research Assistant <i>Institute of Endemic Diseases, Khartoum, Sudan</i> • Supported laboratory and field-based research through sample preparation, reagent handling, data collection, and experimental coordination. • Assisted senior researchers during data acquisition and execution of established laboratory protocols.

Publications and Manuscripts

2026	Safaa Mamoun , Anabela Palandri, Audrey Franceschi Biagioni, Giacomo Amerio, Anna Burato, Giuseppe Legname, Davide Zoccolan, Laura Ballerini, Giada Cellot. <i>Age-Dependent Effects of Social Isolation on Anxiety, Social Behavior, and Neural Substrates in Zebrafish</i> . Manuscript in preparation.
2023	Safaa Mamoun Abdelmageid et al. <i>Association of COMT Val158Met Polymorphism with Fibromyalgia in Khartoum State, Sudan</i> . <i>Pain Research and Management</i> . DOI: 10.1155/2023/7313578
2020	Ibrahim-Elkhalil Abdulrazig et al. <i>Novel Algorithms for PFGE Bacterial Typing: Linking PFGE to WGS Results and Computer Simulations for Evaluation of PulseNet International Typing Protocols</i> . <i>bioRxiv preprint</i> . DOI: 10.1101/2020.07.05.188623

Conference Presentations

2026	Age-Dependent Effects of Social Isolation on Anxiety, Social Behavior, and Neural Substrates in Zebrafish. Poster presentation (upcoming), Computational and System neuroscience meeting (INCTN + TEX Meeting) at SISSA, Trieste, Italy.
2026	Age-Dependent Effects of Social Isolation on Anxiety, Social Behavior, and Neural Substrates in Zebrafish. Poster presentation, FENS Forum, Barcelona, Spain.
2025	Age-Dependent Effects of Social Isolation on Anxiety, Social Behavior, and Neural Substrates in Zebrafish. Oral presentation, TYAN-SISSA Workshop on Emerging Themes in Neuroscience, Trieste, Italy.
2018	Association of COMT Val158Met Polymorphism with Fibromyalgia in Khartoum State, Sudan. Oral presentation, Genomics and Human Health in Africa – Young Scientist Conference.

Training and Summer Schools

2025	Machine Learning Specialization; Mathematics for Machine Learning Specialization, Coursera.
2024	Cajal Advanced Computational Neuroscience Summer School, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2023	Cajal Experimental Neuroscience Bootcamp, SISSA, Trieste, Italy.
2018	Human Brain and EEG Workshop, Sapien Labs, Sudan.
2017	Introduction to Bioinformatics, Pan-African Bioinformatics Network (H3ABioNet), Sudan.

Teaching Experience

2019–2022	University Lecturer <i>Ahfad University for Women, Sudan</i>
2018	Teaching Assistant <i>Pan-African Bioinformatics Network (H3ABioNet), Sudan</i>
2015–2017	University Teaching Assistant <i>University of Khartoum, Sudan</i>

Outreach

Sep 2025	Trieste Next – contributed to exhibition preparation and represented SISSA at the Trieste Science Fair.
Sep 2023	Trieste Next – represented SISSA at the Trieste Science Fair.

Computational and Experimental Skills

Programming: Python, R

Behavioural analysis: DeepLabCut pose estimation; Keypoint-MoSeq; behavioural quantification; supervised classification.

Statistical and computational analysis: entropy-based measures; factor analysis; transition matrices; Markov-chain modelling; machine learning.

Experimental techniques: in-vivo calcium imaging, PCR, Western blot, immunofluorescence, light and confocal microscopy, DNA/protein extraction, histological staining, and animal dissection.

Fellowships and Awards

Sep 2022	SISSA Pre-Doctoral Fellowship – Neurobiology.
Mar 2022	TWAS–SISSA Research Fellowship.

Languages

Arabic (native); English (C1, IELTS 7.5); Italian (intermediate).

References

Prof. Laura Ballerini SISSA, Trieste, Italy ballerin@sisssa.it

Prof. Davide Zoccolan SISSA, Trieste, Italy zoccolan@sisssa.it